

Raghav Mishra

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EDUCATION

SAGE University, Indore

B.Tech (Hons.) in Computer Technology (Artificial Intelligence)

2024–2028

CGPA: 8.02

Jawaharlal Nehru School, Bhopal

Class XII: 75.4% — Class X: 88.0%

2020–2023

SKILLS

- **Languages:** Python, C, C++, JavaScript, SQL, NoSQL, LaTeX
- **Machine Learning:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, XGBoost
- **Deep Learning:** ANN, CNN, NLP, Transformers, Large Language Models (LLMs)
- **Frameworks:** PySpark, Scikit-Learn, PyTorch, TensorFlow, Keras, Node.js
- **Databases:** MySQL, PostgreSQL, MongoDB **Cloud:** Azure, AWS, GCP
- **Core Concepts:** DBMS, Data Structures & Algorithms, AI/ML Architecture

EXPERIENCE

Machine Learning Engineer Intern, MindSynEvolution Pvt. Ltd. , Mumbai

2026–Present

- Built a Streamlit-based UI wrapper over a custom PySpark and AWS Glue job orchestration framework, enabling low-code authoring and automated data pipeline management for enterprise data teams.
- Architected a dual-inference backend: locally powered by an Ollama-served SLM for development and testing, and promoted to production via an AWS Bedrock-integrated Claude model deployed as a Claude Skill for scalable, secure inference.
- Engineered a configuration-driven LLM orchestration layer with intent resolution, dynamic prompt composition, and contract-based output validation to reliably automate pipeline generation workflows.
- Packaged and distributed the solution as an internal Git repository, enabling employees to automate data engineering tasks and reduce manual pipeline coding overhead at scale.

Open Source Contributor, BugViper

github.com/MabudAlam/BugViper

2026

- Identified and fixed a critical thread-safety bug (PR #15, merged) in `src/common/tree_sitter_manager.py` — removed global `Parser` cache causing race conditions under concurrent requests in multi-worker `uvicorn` deployments.
- Refactored `_get_lang_parser()` to instantiate a fresh `Parser` per call while caching only immutable `Language` objects, eliminating shared mutable state across threads.
- Contributed to GSSoC 2026 open-source project BugViper — an AI-powered code review platform built on Neo4j knowledge graphs, LangGraph agents, and Tree-sitter AST parsing across 17 languages.

PATENTS

Electronic Surveillance System and Method of Operating the Same

2026

Indian Patent Application No.: 202621007402 A — Published: 13 March 2026

- Developed an intelligent surveillance architecture integrating computer vision for real-time anomaly detection.
- Designed system framework for automated threat identification using edge and centralized processing.

Wearable Safety Helmet System and Method for Real-Time Threat Assessment

2026

Indian Patent Application No.: 202621003592 A — Published: 27 February 2026

- Designed an AI-enabled safety helmet with environmental sensors for real-time hazard detection in mines.
- Developed hybrid ML architecture and edge-computing inference pipeline for low-latency risk classification.

PROJECTS

DETR — Detection Transformer for Object Detection

2026

github.com/Raghav0079/Detection-Transformer

- Implemented DETR combining CNN backbone with Transformer encoder-decoder for end-to-end detection.
- Eliminated hand-crafted components (NMS/Anchors) using bipartite matching for direct set prediction.

LLM Finetuner — Parameter-Efficient LLM Training Framework

2025

github.com/Raghav0079/LLM-Finetuner

- Built modular framework for fine-tuning open-source LLMs using the Unsloth framework.
- Designed Gradio interface for dataset management and hyperparameter tuning in low-VRAM environments.

CardioPredict Pro — AI-Based Cardiovascular Risk Assessment

2025

github.com/Raghav0079/cardiopredict-pro

- Developed ensemble ML models (Random Forest, SVM, Gradient Boosting) for medical risk prediction.
- Built interactive interface with PostgreSQL backend for real-time probability visualization and reporting.

CERTIFICATIONS

- AWS Machine Learning for Natural Language Processing, Amazon Web Services (2026)
- AWS Data Engineering, Amazon Web Services (2025)
- Microsoft Certified: Azure Fundamentals, Microsoft (2025)
- Google Cloud Engineering Certificate, Google (2025)